SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO5	**Assessment:**	Assessment Tool 1 – **Research Paper Review / Literature Survey**
Weightage:	20%	**Total Marks:**	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	P. MUNI KARTHIK	**Reg. No.:**	192425061
Date:		**Duration:**	60 minutes

Code of Conduct

I P. MUNI KARTHIK / 192425061 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: P. MUNI KARTHIK

Annexure A – Research Paper Review / Literature Survey Instructions

to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem.

The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.

5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing

Future Directions & Relevance(15 %)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation, Citations & References(15 %)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	**Mark**
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

1. Introduction

A Data Warehouse Management System is a centralized analytical environment that collects, integrates, cleans, stores and organizes historical data from multiple sources. It supports online analytical processing (OLAP), reporting, dashboards and data-driven decision making. The purpose of this literature survey is to review existing research related to ETL, multidimensional modelling, requirement analysis, OLAP, data integration, security, semantic integration and modern lakehouse architectures.

The review follows the assignment requirements by selecting relevant research papers, summarizing their objectives, methodologies, technologies and findings, comparing the approaches, identifying limitations and research gaps, and proposing future directions.

2. Selected Research Papers

The following papers were selected because they address important aspects of data warehouse management:

1. A Proposed Model for Data Warehouse ETL Processes – 2011
2. Hierarchy Classification for Data Warehouse: A Survey – 2012
3. User Requirement Analysis in Data Warehouse Design: A Review – 2013
4. Textual Aggregation Approaches in OLAP Context: A Survey – 2017
5. QETL: An Approach to On-Demand ETL from Non-Owned Data Sources – 2017
6. DOLAP Data Warehouse Research over Two Decades: Trends and Challenges – 2019
7. Exploring the Research Landscape of Data Warehousing and Mining Based on DaWaK Conference Full-Text Articles – 2021
8. Incorporation of Ontologies in Data Warehouse/Business Intelligence Systems – A SystematicLiterature Review – 2022
9. Data Lakehouse: A Survey and Experimental Study – 2025

3. Paper 1 – A Proposed Model for Data Warehouse ETL Processes

Year: 2011

Main Area: ETL process management

Objective

The paper focuses on the Extract, Transform and Load (ETL) process, one of the most important components of a data warehouse management system. ETL is complex because data must be extracted from different sources, cleaned, transformed and integrated before being loaded into the warehouse.

Methodology

The research examines approaches for modelling ETL processes, including mapping expressions, guidelines, conceptual modelling and UML-based modelling. It proposes an improved conceptual model for ETL process representation.

Technologies / Concepts

ETL, Data Warehouse, Data Mart, OLAP, Operational Data Store, Data Staging Area, Change Data Capture and Slowly Changing Dimensions.

Major Finding

The paper identifies the absence of a universally accepted standard model for representing ETL scenarios and proposes a conceptual approach to improve ETL design.

Limitation

The research mainly concentrates on ETL modelling and does not completely address modern cloud-scale, streaming and AI-driven warehouse requirements.

Relevance to Data Warehouse Management System

The findings are directly relevant because a Data Warehouse Management System requires an efficient ETL pipeline to collect and integrate data from multiple sources.

4. Paper 2 – Hierarchy Classification for Data Warehouse: A Survey

Year: 2012

Main Area: Dimension hierarchies and multidimensional design

Objective

The research studies dimension hierarchies in data warehouses. For example, Location can be represented as City → State → Country. Such hierarchies allow users to analyse data at different levels of detail.

Methodology

The authors survey different hierarchy classifications proposed by researchers and compare them based on relevant parameters.

Technologies / Concepts

Multidimensional schema, dimension, dimension hierarchy, OLAP, roll-up and drill-down.

Major Finding

Proper hierarchy classification is important for designing flexible data warehouse systems capable of supporting different levels of analysis.

Limitation

The work focuses mainly on hierarchy classification and does not provide a complete solution for the entire data warehouse management lifecycle.

Relevance to Data Warehouse Management System

A Data Warehouse Management System requires properly designed dimensions and hierarchies to support efficient analytical queries.

5. Paper 3 – User Requirement Analysis in Data Warehouse Design: A Review

Year: 2013

Main Area: Requirement analysis in data warehouse design

Objective

The paper investigates user requirement analysis during data warehouse design. Requirement analysis affects important decisions about what information the warehouse should contain and how it should support business intelligence.

Methodology

The research categorizes requirement-analysis approaches into data-driven, user-driven, goal-driven and mixed-driven approaches.

Technologies / Concepts

Data-driven, user-driven, goal-driven and mixed-driven requirement methods.

Major Finding

The research shows that requirement analysis influences almost every decision in a data warehouse or BI implementation and that no single universally standardized approach is suitable for every project.

Limitation

Selecting the appropriate requirement-analysis method can be difficult for data warehouse designers.

Relevance to Data Warehouse Management System

Before developing a Data Warehouse Management System, requirements must be collected from managers, analysts and other users.

6. Paper 4 – Textual Aggregation Approaches in OLAP Context: A Survey

Year: 2017

Main Area: Textual data analysis in OLAP

Objective

Traditional OLAP systems are particularly effective for numerical data. This research examines methods for handling textual data in OLAP environments.

Methodology

The paper surveys textual aggregation methods and classifies approaches into cube-structure-based and text-mining-based groups. It also considers textual similarity measures and datasets used for textual aggregation.

Technologies / Concepts

OLAP, text mining, textual similarity and aggregation.

Major Finding

Traditional numerical aggregation techniques are insufficient for large amounts of textual information.

Limitation

Textual data is more difficult to aggregate and analyse than structured numerical data.

Relevance to Data Warehouse Management System

A modern Data Warehouse Management System may need to process customer reviews, feedback, social media information, text documents and support tickets.

7. Paper 5 – QETL: An Approach to On-Demand ETL from Non-Owned Data Sources

Year: 2017

Main Area: Reducing the cost of unnecessary ETL

Objective

The paper proposes Query-Extract-Transform-Load (QETL) for situations where loading all available data into a warehouse beforehand is expensive or impractical.

Methodology

Instead of loading everything in advance, the required data is identified from an analytical query, extracted, transformed and loaded into an analytical cube. Data can be removed when storage is required.

Technologies / Concepts

On-demand ETL, OLAP cubes and a genomics case study.

Major Finding

On-demand extraction can reduce unnecessary extraction costs and improve performance through data reuse.

Limitation

The approach is particularly useful for specific scenarios and does not replace every traditional ETL architecture.

Relevance to Data Warehouse Management System

It provides an alternative approach for managing very large external datasets.

8. Paper 6 – DOLAP Data Warehouse Research over Two Decades: Trends and Challenges

Year: 2019

Main Area: Evolution of data warehouse research

Objective

This research examines the evolution of data warehouse research over approximately two decades.

Methodology

The study analyses research trends and changes in the topics investigated in data warehousing and analytical processing.

Technologies / Concepts

DOLAP and research topics associated with VLDB, SIGMOD and ICDE communities.

Major Finding

Earlier research focused heavily on relational technologies, while the growth of big data shifted attention toward non-relational systems and broader analytical technologies.

Limitation

The paper is primarily a research-trend analysis rather than a complete warehouse implementation framework.

Relevance to Data Warehouse Management System

It demonstrates why a modern Data Warehouse Management System must evolve beyond traditional relational warehouse architectures.

9. Paper 7 – Exploring the Research Landscape of Data Warehousing and Mining Based on DaWaK Conference Full-Text Articles

Year: 2021

Main Area: Research trends in data warehousing and mining

Objective

This study investigates research trends in data warehousing and data mining using papers from the DaWaK conference.

Methodology

The researchers used co-word analysis, topic modelling, co-author network analysis and network visualization.

Technologies / Concepts

Conference research papers, text mining, topic modelling and network analysis.

Major Finding

Important research topics include data mining, algorithm performance, information systems, database encryption and OLAP. Security-related topics such as database encryption became increasingly important.

Limitation

The analysis is based on a research corpus rather than a specific complete warehouse implementation.

Relevance to Data Warehouse Management System

Security and advanced analytics should be important components of a modern Data Warehouse Management System.

10. Paper 8 – Incorporation of Ontologies in Data Warehouse/Business Intelligence Systems

Year: 2022

Main Area: Semantic integration and data heterogeneity

Objective

This systematic literature review investigates how ontologies can be incorporated into Data Warehouse and Business Intelligence systems.

Methodology

The research reviews ontology-based solutions for dimensional modelling, requirement analysis, ETL and BI application design.

Technologies / Concepts

Ontologies, semantic integration, ETL, BI and interoperability concepts.

Major Finding

Semantic technologies can improve the integration and interpretation of heterogeneous data.

Limitation

Ontology-based architectures can introduce additional modelling complexity.

Relevance to Data Warehouse Management System

A Data Warehouse Management System integrating multiple departments and heterogeneous sources could benefit from semantic metadata.

11. Paper 9 – Data Lakehouse: A Survey and Experimental Study

Year: 2025

Main Area: Large-scale heterogeneous data management

Objective

The research examines the evolution from traditional data warehouses and data lakes toward data lakehouse architectures.

Methodology

The study compares data lakes, data warehouses and data lakehouses and evaluates systems including HDFS, Hive and Delta Lake using analytical queries and an IMDb dataset.

Technologies / Concepts

HDFS, Hive, Delta Lake, IMDb dataset and analytical query processing.

Major Finding

A lakehouse combines several advantages of data lakes and warehouses and is designed to handle structured and unstructured data at large scale.

Limitation

Lakehouse architectures can introduce new challenges in governance, architecture and system management.

Relevance to Data Warehouse Management System

The research is important for the future development of Data Warehouse Management Systems handling big data and unstructured information.

12. Comparative Literature Survey

The comparative survey below is provided as an image instead of a manually formatted PDF table, as requested. It contains the research problem, methodology, tools/data, findings and limitations for the selected studies.

Paper / Year	Research Problem	Methodology	Tools / Data	Major Finding	Limitation
Proposed Model for DW ETL	ETL complexity	Conceptual ETL modelling	ETL, UML, DW concepts	Improved ETL modelling	Limited modern/cloud focus
Hierarchy Classification / 20	Dimension hierarchy manag	Survey and comparison	OLAP, multidimensional sch	Better hierarchy classificati	Focuses mainly on hierarchies
User Requirement Analysis	DW requirement identificati	Review of approaches	Data/user/goal-driven meth	Requirement analysis is crit	No universal approach
Textual Aggregation in OLAI	Text data analysis	Literature survey	Text mining, OLAP	Text needs specialized aggr	Text aggregation complexity
QETL / 2017	Expensive full ETL	On-demand ETL	OLAP cube, genomics case	Reduces unnecessary extra	Scenario-specific
DOLAP Trends / 2019	Evolution of DW research	Research trend analysis	DOLAP / VLDB / SIGMOD / I(	Shift toward big data/non-re	Trend study rather than implem
DaWaK Landscape / 2021	Research evolution	Text mining and topic mode	Conference papers	Security and data mining g	Based on research corpus
Ontologies in DW/BI / 2022	Data heterogeneity	Systematic literature reviev	Ontologies, OWL, ETL, BI	Improves semantic integrat	Added modelling complexity
Data Lakehouse / 2025	Large-scale heterogeneous	Survey + experiment	HDFS, Hive, Delta Lake, IMD	Lakehouse combines wareh	Governance and architecture c

Figure 1. Comparative literature survey table.

13. Limitations and Challenges of Existing Approaches

1. ETL Complexity

ETL requires data extraction, cleaning, transformation and integration from multiple sources. It remains a major and complex part of warehouse implementation.

2. Data Quality

Data may contain missing values, duplicate records, incorrect formats, inconsistent values and invalid records. Poor data quality affects analytical results.

3. Heterogeneous Data Sources

Organizations may have data from relational databases, CSV files, APIs, enterprise applications, cloud systems and external sources. Integrating these sources remains challenging.

4. Scalability

Traditional architectures may face performance problems as data volume increases.

5. Complex OLAP Queries

Large analytical queries can require significant processing and storage resources.

6. Unstructured Data

Traditional warehouses are primarily designed for structured information, while textual and other unstructured data require additional processing techniques.

7. Data Security

Data warehouses contain valuable organizational information, so authentication, access control, encryption and auditing are important.

8. Requirement Uncertainty

Different users may have different analytical requirements, making requirement analysis difficult.

9. Semantic Heterogeneity

Different data sources may use different names and meanings for similar information, creating integration problems.

14. Research Gap

Research Gap 1 – Integrated Data Warehouse Management

Many studies focus on individual components such as ETL, OLAP, hierarchies, requirements and semantic integration. There is a need for an integrated Data Warehouse Management System that combines these functions into one manageable architecture.

Research Gap 2 – Real-Time Data Processing

Traditional warehouses commonly rely on scheduled ETL processes. There is a need for systems supporting both batch processing and near real-time processing.

Research Gap 3 – Heterogeneous Data Integration

Modern organizations generate data from structured and unstructured sources. A future system should integrate SQL databases, CSV files, APIs, cloud data and textual data efficiently.

Research Gap 4 – Intelligent ETL

Existing ETL approaches require substantial configuration and development effort. AI/ML can assist with anomaly detection, duplicate detection, transformation recommendations and data-quality monitoring.

Research Gap 5 – Scalable Architecture

The research trend toward big data and non-relational systems indicates a need for distributed, parallel and cloud-oriented warehouse architectures.

Research Gap 6 – Unified Warehouse and Lake Architecture

The development of lakehouse architectures demonstrates the need to manage structured and unstructured data together. Integrating traditional warehouse capabilities with data-lake capabilities is an important future direction.

15. Possible Improvements and Future Research Directions

1. AI-Based ETL

Develop an intelligent ETL system capable of automatically detecting data-quality problems and recommending transformations.

2. Real-Time Data Warehouse

Introduce streaming technologies so that new data can be incorporated continuously rather than only through scheduled batch processing.

3. Cloud-Based Data Warehouse

Use cloud infrastructure to provide elastic scalability, high availability, distributed storage and flexible computing resources.

4. Automated Data Quality Management

Automatically detect missing data, duplicate data, outliers and inconsistent formats.

5. Advanced Security

Incorporate role-based access control, encryption, authentication, audit logs and fine-grained data access.

6. Semantic Data Integration

Use ontology-based methods to improve interoperability between heterogeneous data sources.

7. Lakehouse Integration

Combine data-lake and data-warehouse capabilities to support structured and unstructured information.

8. Predictive Analytics

Integrate machine-learning models for sales prediction, customer behaviour analysis, demand forecasting, risk prediction and fraud detection.

16. Proposed Data Warehouse Management System Workflow

The proposed workflow integrates data sources, ingestion, ETL/ELT, staging, warehouse storage, OLAP and decision-support outputs. The workflow below is provided as an image instead of a manual drawing, as requested.

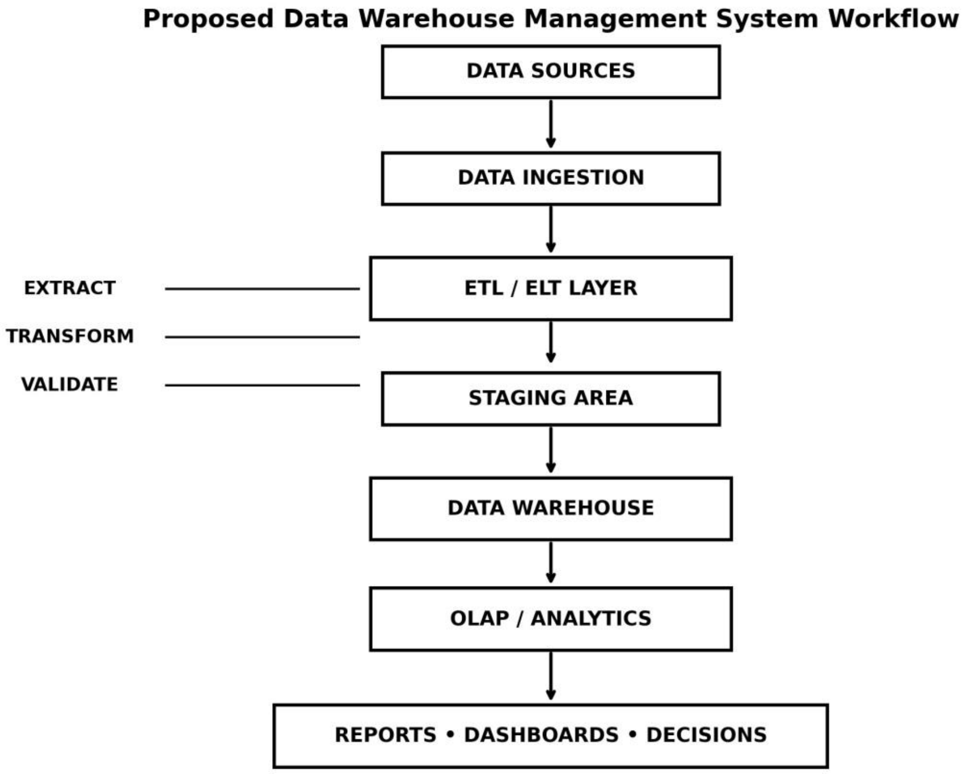

Figure 2. Proposed Data Warehouse Management System workflow.

17. Main Components of the Proposed System

1. Data Sources: Collect data from multiple operational systems such as relational databases, CSVfiles and APIs.
2. Data Ingestion: Collect and transfer source data into the warehouse environment.
3. ETL/ELT Layer: Extract, transform, clean, validate and load the data.
4. Staging Area: Temporarily store data before loading it into the warehouse.
5. Data Warehouse: Store integrated historical data for analysis.
6. Fact Tables: Store measurable business information such as sales amount and quantity.
7. Dimension Tables: Store descriptive information such as customer, product, time and location.
8. OLAP Layer: Support roll-up, drill-down, slice, dice and pivot operations.
9. Reporting and Dashboard Layer: Present information to managers, analysts, executives anddecision makers.

18. Example Warehouse Schema

Sales_Fact: Sales_ID, Product_ID, Customer_ID, Time_ID, Sales_Amount, Quantity

Dimension Tables: Customer Dimension, Product Dimension, Time Dimension and Location Dimension.

19. Conclusion

The literature review shows that Data Warehouse Management Systems have evolved from traditional relational warehouses and periodic ETL processes toward more scalable, heterogeneous and intelligent data-management architectures. The reviewed studies demonstrate that ETL design, requirement analysis, multidimensional modelling, OLAP, data integration, security and scalability are important components of data warehouse systems.

20. References

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4. M. Bouakkaz, Y. Ouinten, S. Loudcher, and Y. Strekalova, “Textual Aggregation Approachesin OLAP Context: A Survey,” International Journal of Information Management, vol. 37, no. 6, pp. 684–692, 2017.

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